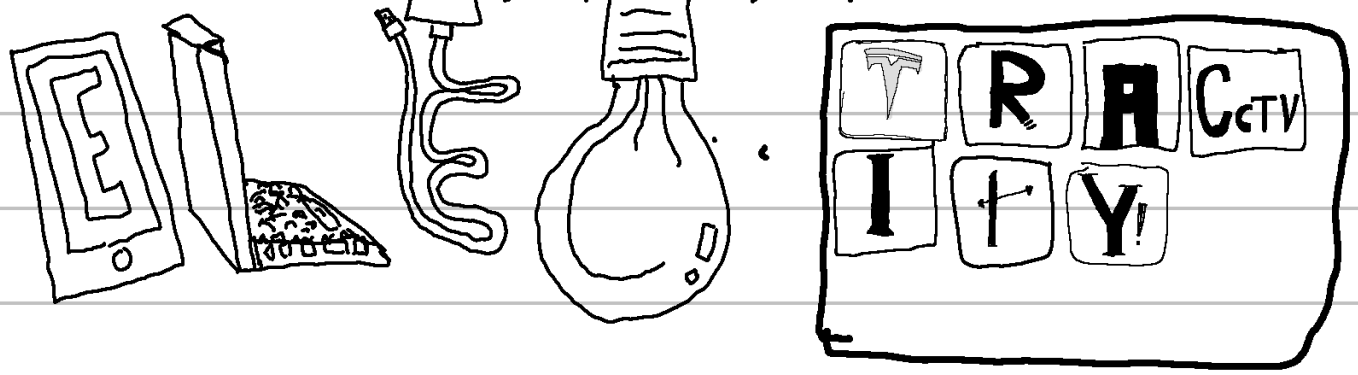
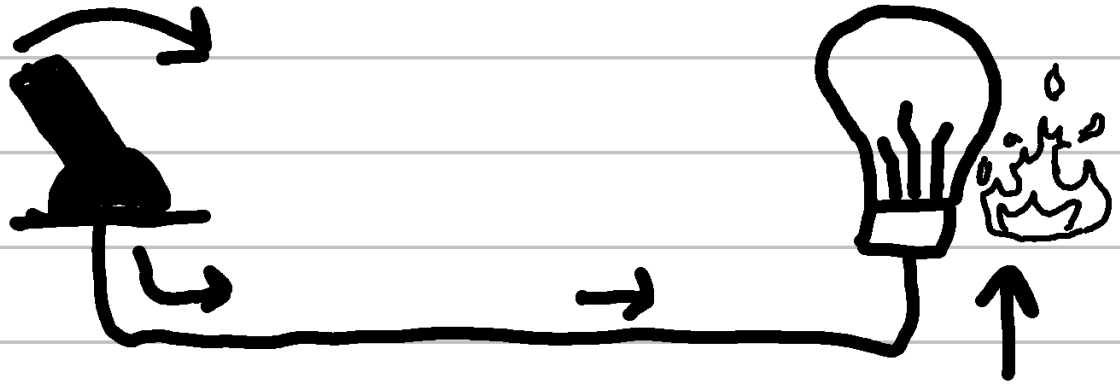


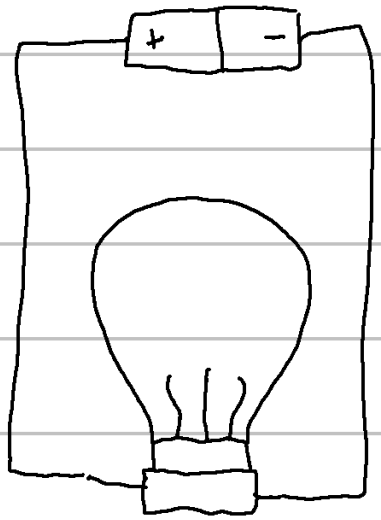


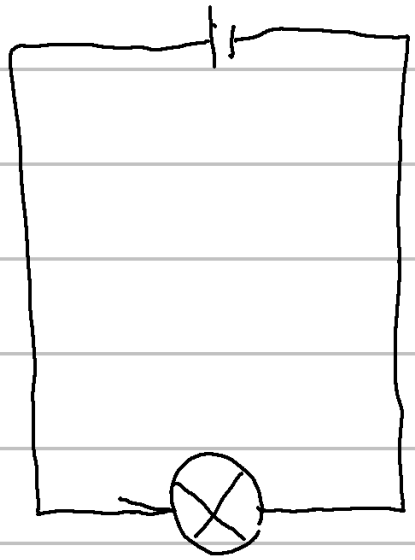
- Electricity is a form of energy.
- Electrons move through materials.
- Batteries push electrons.
- Electricity needs a path.

CIRCUITS

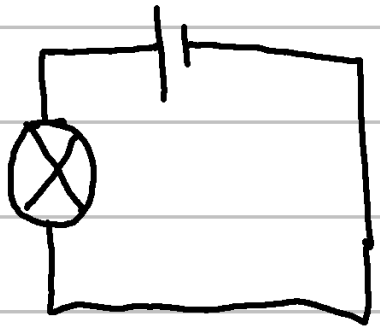
components of an electrical circuit

- Power source.
- wires.
- components.



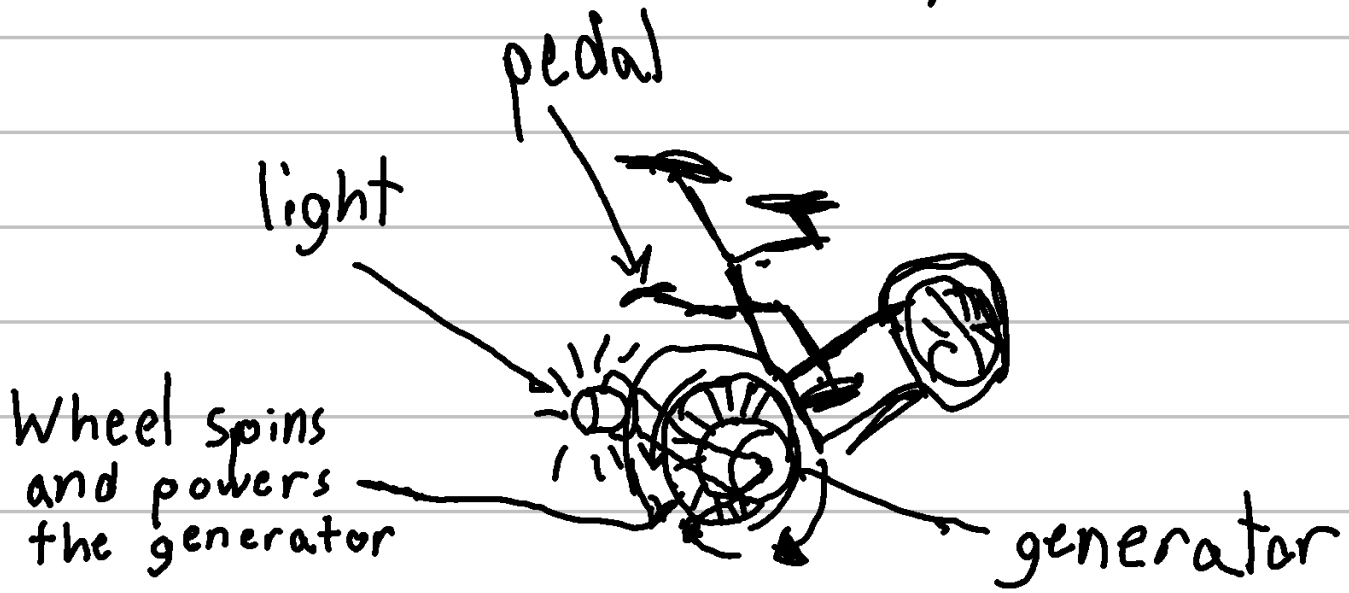


- a circuit is a pathway for electricity
- circuits let us control electricity
- circuits power devices.
- circuits must be complete to work.



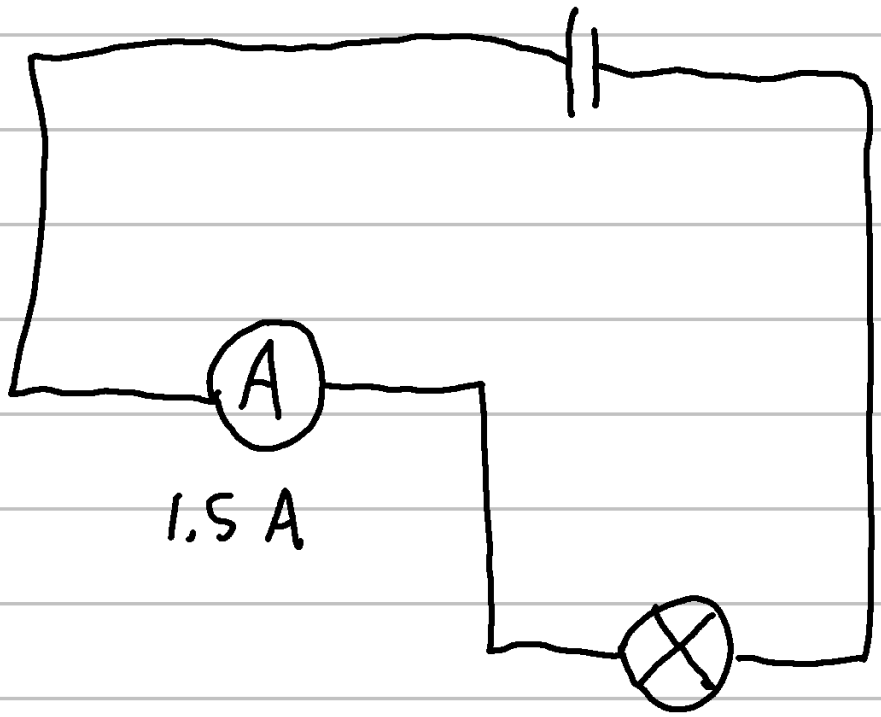
a circuit is a loop.

Bill Nye

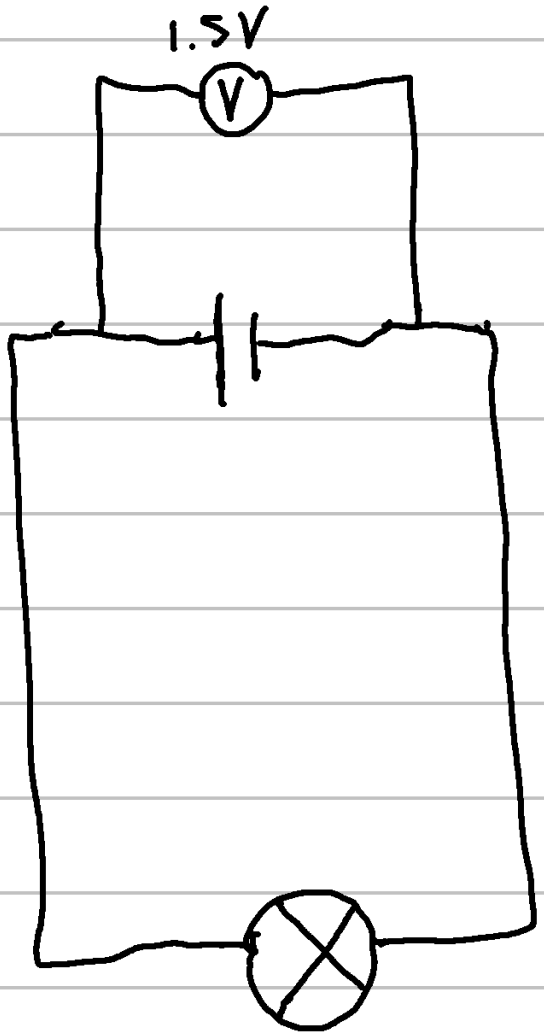


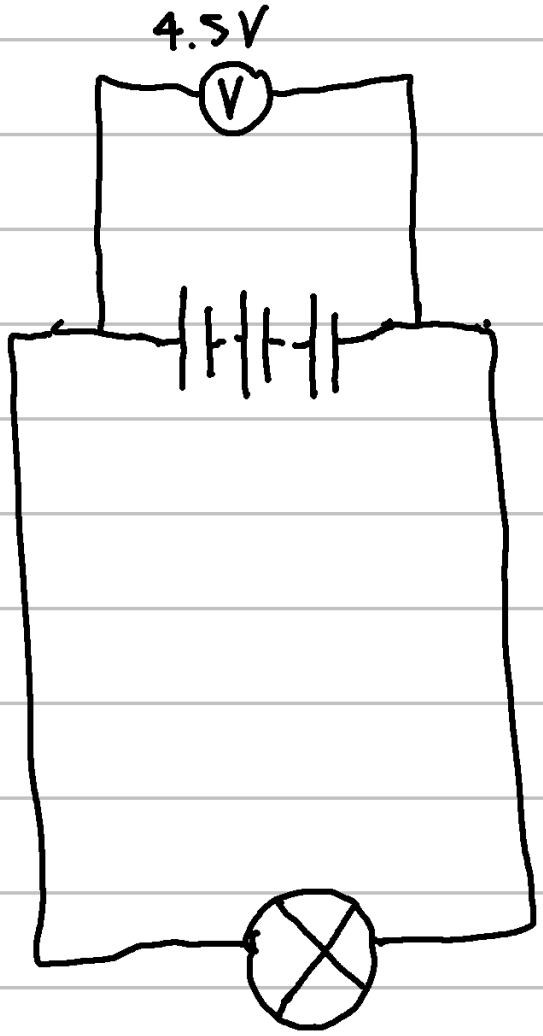
- electricity needs a complete path.
- switches open/close the path.
- electricity powers machines.
- real world examples: toaster, light, motor, tv, gameboy, Grandpa.

- The units of current are called amperes, often shortened to Amps. These are the flow of the electrons in a wire.
- measured with an ammeter.
- ammeters can be placed anywhere in a circuit.
- 1.5 amps can be written as 1.5 A.



- The unit of potential difference is the volt. That is the amount of energy transfered by each unit of charge
- measured with a voltmeter.
- The voltmeter is used in a parallel circuit, it cannot be used in a series circuit.
- 1.5 volts can be written as 1.5 V.





essential parts that make up a circuit

Power source: energy provider, ex. battery, power outlet. Give electrons the push they need to move.

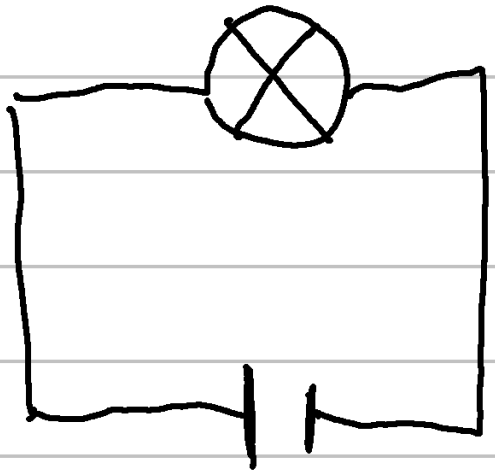
conductors: pathways, ex. metal or copper wire. Facilitates the flow of electrons.

device: component, ex. bulb, fan, motor. uses electricity.

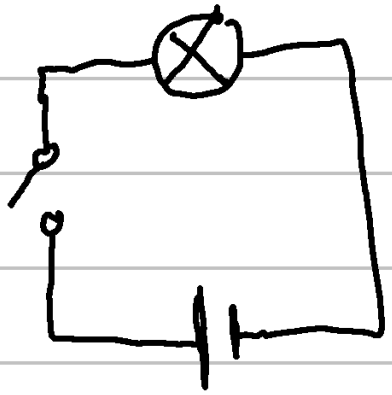
switch: gives you control.
open circuit = turned off.
closed circuit = turned on.

Drawing circuits with given prompts

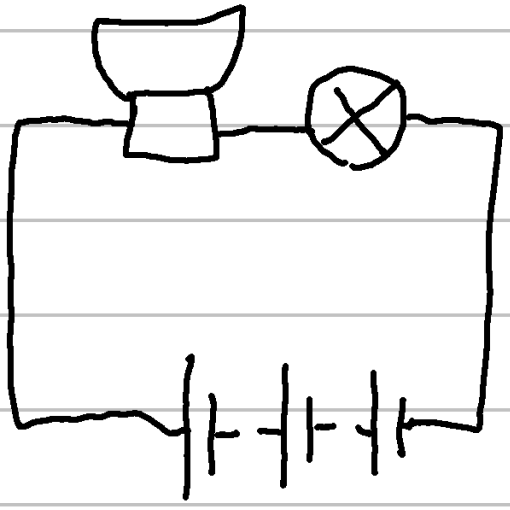
1)



2)



3)



Conductors and Insulators

conductors : materials that electricity can pass through easily, ex. Metal, graphite.

metals have lots of free electrons.

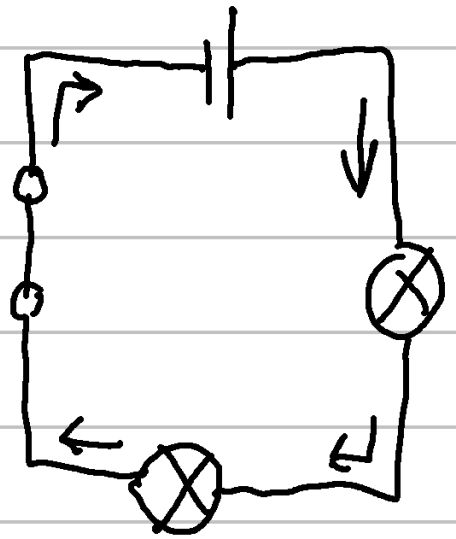
insulators : materials that electricity cant pass through easily, ex. plastic, glass, rubber.

tap water, salt water, and rain water are good conductors and electrolytes

gases arent generally good conductors, except when lightning strikes.

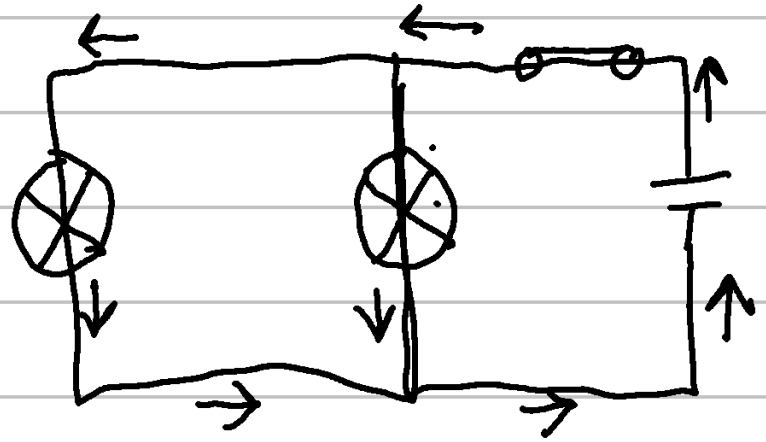
Series Circuits

- Series circuit have one continuous pathway
- if on bulb goes out → all bulbs go out.
- bulbs get dimmer as you add more → they share electrical energy (Voltage)



Parallel Circuits

- parallel circuits have more than 1 path (branches)
- if one component goes out → the other(s) stay on.
- bulbs each hold the full electrical energy (voltage)
  - stays bright



series	parallel
one path	many paths
if the bulb breaks they all turn off	if the bulb breaks they all stay on
voltage - energy is shared.	voltage - all components get a full amount of energy.
resistance: adding more parts makes it harder for electricity to flow	resistance: adding more parts makes it easier for electricity to flow

Electricity and circuits vocab

Electricity: the flow of electrons that carry energy, ex. the power that makes the lights turn on.

Electrons: negatively charged part in an atom that are always moving, they create electricity.

Current: the measure of how many electrons are moving through the circuit each second. like the "volume" of electricity.

voltage: the force that pushes electrons through a circuit

battery: a power source that turns chemical energy into electrical energy to provide the push to make electrons move.

conductor: a material that allows electricity to flow through easily, ex. metal, graphite.

insulator: a material that doesn't allow electricity to pass through easily, ex. plastic, wood, rubber.

circuit: an unbroken loop that starts at a power source and eventually returns to it.

closed circuit: an unbroken loop that starts at a power source goes through all the components and eventually returns to it, ex. turning on a flashlight

Open circuit: a circuit with a break in the path preventing electricity from flowing.

Switch: can open or close a circuit, controlling the flow of electricity.

load: Any device in a circuit that uses electrical energy an converts it into another form (light, heat, sound, motion) ex. bulb, speaker, motor.

Resistance: Anything in a circuit that slows down or opposes the flow of electrons turning electrical energy into another form.

Series circuit: a circuit with only one path for energy to flow, if one component breaks, everything stops.

parallel circuit: a circuit with more than one path for electricity to flow, if a component breaks the others keep working.

Ammeter: a tool used to measure the electric current flowing through a circuit.

Voltmeter: A tool used to check the voltage (electrical pressure) between two points in a circuit.

Safety

never touch bare wires

keep water away

always ask an adult

use proper equipment

always treat electricity with respect